Homework 09

REAL VARIABLES I

Fall 2026

Due: 2026-11-08 19:00
Topics: Density of C_c in $L^p(\mathbb{R}^d)$, L^p norms as $p \to \infty$, continuity of translations in $L^p(\mathbb{R})$, sets of exponents p with $f \in L^p$.

Instructions

Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

---

1. (Density of compactly supported continuous functions)

Let μ be a Borel measure on $\mathbb{R}^d$ with $\mu(K) < +\infty$ for every compact set $K \subseteq \mathbb{R}^d$, and let $p \in [1, +\infty)$. Denote by $C_c(\mathbb{R}^d)$ the continuous functions $g \colon \mathbb{R}^d \to \mathbb{R}$ whose support $\operatorname{spt} g = \overline{\{g \neq 0\}}$ is compact.

1.a

weight: 1.0 Prove that $C_c(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d, \mu)$.

> **Free hint.** First cut by $\mathbb{1}_{B_R(0)}$ for some large R. Then approximate an function supported on a finite radius ball.

2. (L^p converge to L^∞ as $p \to \infty$)

Let $(X, \mathcal{E}, \mu)$ be a measure space and let $f \colon X \to \mathbb{R}$ be measurable.

2.a

weight: 0.7 Prove that $\liminf_{p\to\infty} \|f\|_{L^p} \geq \|f\|_{L^\infty}$, and that if $\|f\|_{L^{p_0}} < +\infty$ for some $p_0 \in [1,\infty)$ then

$$\lim_{p\to\infty} \|f\|_{L^p} = \|f\|_{L^\infty} \qquad \text{in } [0,+\infty].$$

Address the case when the quantities in play are infinite (or discuss why that cannot be the case).

2.b

weight: 0.3 Show that the assumption $\|f\|_{L^{p_0}} < +\infty$ for some p_0 cannot be dropped: give a measure space and a measurable f with $\|f\|_{L^\infty} < +\infty$ for which $\lim_{p\to\infty} \|f\|_{L^p} \neq \|f\|_{L^\infty}$.

3. (Continuity of translations)

IN CLASS

Let $p \in [1,\infty)$, and equip $\mathbb{R}$ with Lebesgue measure $\mathcal{L}$. You may use translation invariance of Lebesgue measure in the form $\int_{\mathbb{R}} g(x-h)\,\mathrm{d}x = \int_{\mathbb{R}} g(x)\,\mathrm{d}x$ for every measurable $g\colon \mathbb{R} \to [0,+\infty]$ and $h \in \mathbb{R}$. Equip $\mathbb{R} \times L^p(\mathbb{R})$ with the product distance

3.a

weight: 0.5 Prove that the map

$$\mathbb{R} \times L^p(\mathbb{R}) \to L^p(\mathbb{R}), \qquad (h,f) \mapsto f(\cdot - h)$$

is continuous.

> **Free hint.** First approximate f by a good $C_c^0(\mathbb{R})$ function. Then use the triangle inequality multiple times:
>
> $$f(x-h) - f(x) = f(x-h) - g(x-h) + g(x-h) - g(x) + g(x) - f(x)$$

3.b

weight: 0.4 Prove that the map $(h,f) \mapsto f(\cdot - h)$ is not locally uniformly continuous i.e. for any open neighborhood of any point the map fails to be uniformly continuous.

> **Free hint.** Consider functions of the form $\varepsilon 2^{n/p} \mathbb{1}_{[0,2^{-n}]}$

3.c

weight: 0.1 Deduce from (b) that no closed ball $\{f \in L^p(\mathbb{R}) : \|f - f_0\|_{L^p(\mathbb{R})} \leq r\}$ with $r > 0$ is compact in $L^p(\mathbb{R})$. In particular, closed and bounded subsets of $L^p(\mathbb{R})$ need not be compact.

Remark. The conclusion of (c) also follows from a general fact: the closed unit ball of a Banach space is compact if and only if the space is finite-dimensional. Since the functions $\mathbb{1}_{[2^{-k-1},2^{-k})}$, $k \in \mathbb{N}$, are linearly independent in $L^p(\mathbb{R})$, the space $L^p(\mathbb{R})$ is infinite-dimensional, and (c) is immediate. The argument above instead exhibits a concrete mechanism behind the failure of compactness: the

functions $\delta^{-1/p}\mathbb{1}_{[0,\delta)}$ concentrate on smaller and smaller intervals. The continuity in (a) fails for $p = \infty$: $\|\mathbb{1}_{[0,1]}(\cdot - h) - \mathbb{1}_{[0,1]}\|_{L^\infty(\mathbb{R})} = 1$ for every $h \neq 0$.

4. (Prescribing the set of exponents)

Let $d \geq 1$, equip $\mathbb{R}^d$ with Lebesgue measure $\mathcal{L}$, and let $0 < a < b < \infty$. For each of the four intervals

$$(a, b), \qquad [a, b], \qquad [a, b), \qquad (a, b],$$

give a measurable function $f\colon \mathbb{R}^d \to [0, +\infty)$ such that $f \in L^p(\mathbb{R}^d)$ for exactly the exponents $p \in (0, \infty)$ in that interval.

4.a

weight: 1.0 Construct the four functions and sketch why each works.

Free hint.

- Combine negative powers $|x|^{-\alpha}$ and powers of a logarithm, such as $\left(1+\left|\log|x|\right|\right)^{-\beta}$, multiplied by $\mathbb{1}_{B_1(0)}$ or by $\mathbb{1}_{\mathbb{R}^d \setminus B_1(0)}$. The behavior near 0 and near ∞ can be adjusted independently.
- You may use polar coordinates: $\int_{\mathbb{R}^d} g(|x|)\,\mathrm{d}x = c_d \int_0^\infty g(r)\, r^{d-1}\,\mathrm{d}r$ for measurable $g\colon [0, \infty) \to [0, +\infty]$, where $c_d \in (0, \infty)$ depends only on d.